

introduction:

I started this project with some what of an idea bioinformatics is due to a previous project like this with python it was very basic and only generated a random string sequence with no proper meaning where as this project in c++ is alot more complex but efficient and gives off good results.

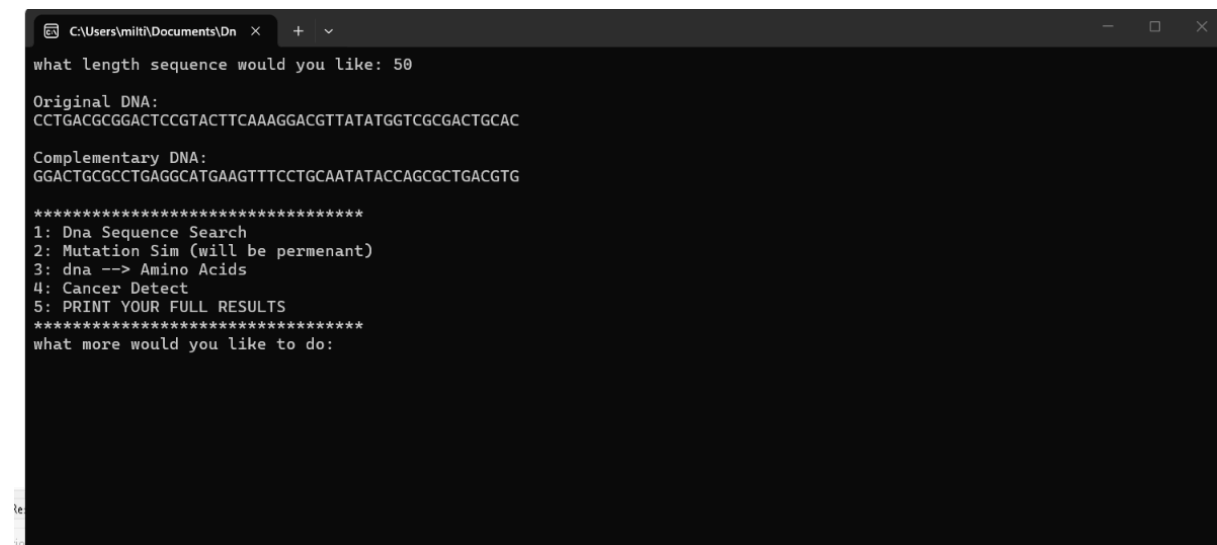
I would consider this project a professional level project that doctors or anyone like that would use but still great for those who want to get generated data easily and can use that data for their own use

Beginning of my project:

This was essentially my most difficult part as i need to make a clear plan and idea of what i was doing but as i did this i had constantly came across new things i could do. Is started off by making my generation, which went well i let the user input the length so that its then personalized to their needs, i also decided that i wanted the Rna to be shown with the original as Dna and Rna are complementary

(<https://study.com/academy/lesson/video/template-dna-definition-lesson-quiz.html>)

I also made the menu which has been changed multiple times due to the change of ideas but thats how we have this:



```
C:\Users\milti\Documents\Dn x + v
what length sequence would you like: 50

Original DNA:
CCTGACGCGGACTCCGTACTTCAAAGGACGTTATATGGTCGCGACTGCAC

Complementary DNA:
GGACTGCGCCTGAGGCATGAAGTTTCCTGCAATATACCAGCGCTGACGTG

*****
1: Dna Sequence Search
2: Mutation Sim (will be permanent)
3: dna --> Amino Acids
4: Cancer Detect
5: PRINT YOUR FULL RESULTS
*****
what more would you like to do:
```

Dna search:

I got this idea because of my research through wikipedia because of codons which are in groups 3 i ultimately assumed that this was the idea so i took my research and checked and i was correct, to actually make it i made a variable (search) which the user will input what they want i then i make a new variable again to search for the users input, i made so that if the search is in the strand it will give the user the position that it is at and if not then it would just give a not found message.

Dna -> Amino acids:

I found this part the most interesting as i learnt the most about genetic code here. But anyways sticking with the wikipedia page

(https://en.wikipedia.org/wiki/DNA_and_RNA_codon_tables#Translation_table_1)

I was able to find out about codons and amino acids, i must admit i was lazy as i could have turned the dna codons into rna codons and then amino acids but I didn't do so. I also did this part before my mutation as i had not found out about mutations yet, but i understood what was in front of me and i was happy to make that as of now:

Amino-acid biochemical properties	Nonpolar (np)	Polar (p)	Basic (b)	Acidic (a)		Termination: stop codon *	Initiation: possible start codon =>
-----------------------------------	---------------	-----------	-----------	------------	--	---------------------------	-------------------------------------

Standard genetic code^{[18][note 3]}

1st base	2nd base								3rd base	
	T		C		A		G			
T	TTT	(Phe/F) Phenylalanine	TCT		TAT	(Tyr/Y) Tyrosine (p)	TGT	(Cys/C) Cysteine (p)	T	
	TTC	(np)	TCC		TAC		TGC		C	
	TTA		TCA		(Ser/S) Serine (p)	TAA	Stop (Ochre) * ^[note 2]	TGA	Stop (Opal) * ^[note 2]	A
	TTG ⇒		TCG		TAG	Stop (Amber) * ^[note 2]	TGG	(Trp/W) Tryptophan (np)	G	
C	CTT	(Leu/L) Leucine (np)	CCT		CAT	(His/H) Histidine (b)	CGT	(Arg/R) Arginine (b)	T	
	CTC		CCC		CAC		CGC		C	
	CTA		CCA		CAA	(Gln/Q) Glutamine (p)	CGA		A	
	CTG		CCG		CAG		CGG		G	
A	ATT	(Ile/I) Isoleucine (np)	ACT	(Thr/T) Threonine (p)	AAT	(Asn/N) Asparagine (p)	AGT	(Ser/S) Serine (p)	T	
	ATC		ACC		AAC		AGC		C	
	ATA		ACA		AAA	(Lys/K) Lysine (b)	AGA	(Arg/R) Arginine (b)	A	
	ATG ⇒		ACG		AAG		AGG		G	
G	GTT	(Val/V) Valine (np)	GCT	(Ala/A) Alanine (np)	GAT	(Asp/D) Aspartic acid (a)	GGT	(Gly/G) Glycine (np)	T	
	GTC		GCC		GAC				GGC	C
	GTA		GCA		GAA	(Glu/E) Glutamic acid (a)	GGA		A	
	GTG ⇒		GCG		GAG		GGG		G	

This list was really handy and was very clear for me to use it to make it into something for my project. I used map to store my data into a list this was also the longest part as there was way to much going back and forth between tabs but we got it i made it so that i had loop that scanned through every 3 letters which then was able to identify what amino acid each of the codons. I only made it that the dna codons are shown but i will later on make it so that we get the rna codons.

Mutation sim:

I found it cool that i could turn this into a possible feature thats when i stumbled across it on wikipedia and from that i got the idea that i would get the user to enter how much of the strand will be mutated and then i would simulate that effect

```
Original DNA:
CCTGACGCGGACTCCGTACTTCAAAGGACGTTATATGGTCGCGACTGCAC
```

```
what more would you like to do: 2
*****
how much percentage of the original dna would you like to mutate: 45
mutation at 41:C --> G
mutation at 34:A --> T
mutation at 24:A --> C
mutation at 8:G --> C
mutation at 14:C --> T
mutation at 45:T --> G
mutation at 11:C --> G
mutation at 45:G --> C
mutation at 27:A --> G
mutation at 4:A --> C
mutation at 3:G --> A
mutation at 32:A --> T
mutation at 16:T --> C
mutation at 45:C --> G
mutation at 26:G --> C
mutation at 19:T --> A
mutation at 17:A --> G
mutation at 35:T --> C
mutation at 3:A --> G
mutation at 22:A --> T
mutation at 23:A --> T
mutation at 11:G --> T
CCTGCCGCCGATTCTGCGCATCTTCGCGCGTTTTTCGGTCGGGACGGCAC
```

I was pretty happy with this its not the most realistic but its only for a bit of messing around, i do intend on making this a more serious project later on with more realism in it

Cancer detect:

This was a very confusing part and i didnt want to get stuck with this yet as i found the mutations hard enough but once i get more skill and a better understanding of it i will definitely include this into my next model of this project

Results:

I actually knew abit about some of the stats given already so it made it easier for me to do this as my intention with this was to give people the gc percentage and at percentage which both us the same formula e.g: $((a + t \text{ amounts}) / \text{total in sequence}) \times 100$

This gave us a very easy percentage which is realised instead of just counting how many individual a's c's t's and g's there was i could also include a percentage for them using the same formula as the atPercentage and gcPercentage

```

*****
1: Dna Sequence Search
2: Mutation Sim (will be permanent)
3: dna --> Amino Acids
4: Cancer Detect
5: PRINT YOUR FULL RESULTS
*****
what more would you like to do: 5
*****
total: 50
-----results-----
adenine(a): 11 (22%)
thymine(t): 11 (22%)
cytosine(c): 15 (30%)
guanine(g): 13 (26%)
gc: 28
gc percentage: 56%
at percentage: 44%
-----

```

I was very happy with this as it look good and accurate and just in general looked like information that could be used.

Overall:

Im happy with what i have done with this project and i feel like i have learned alot with this project not just coding wise but biology wise i do intend on working on this project in the near future and i hope to fix all the bits im not happy with other than that i am very proud of what i have made still and feel like rather than saying I could've done better i should just be happy with whats in front of me and when the time comes i can work on it more.